

# NASSER FENDRI

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## EDUCATION

**M.Sc. Mathematics** — Concentration in Actuarial & Financial Mathematics Jan 2023 – Feb 2025

**Université du Québec à Montréal (UQAM)** GPA: 3.93/4.3 | Excellence Scholarship

Relevant Coursework: Stochastic Methods in Finance · Stochastic Processes · Probability & Measure Theory · Mathematical Risk Analysis · Financial Mathematics · Structured Products · Financial Engineering (incl. CVA/XVA) · Applied Linear Models

**National Engineering Degree** Sep 2019 – Jul 2023

**Tunisia Polytechnic School** Score: 15.49/20

Relevant Coursework: Probability & Statistics · Optimization · Numerical Analysis · Algorithms & Data Structures · Econometrics · Derivatives & FX Markets · Machine Learning · Deep Learning

**Preparatory Institute of Engineering Studies (IPEIS), Sfax** Sep 2017 – Jun 2019

National Competitive Entrance Exam — Ranked 34th / 1,600 candidates nationwide

## TECHNICAL SKILLS

**Programming:** Python (NumPy, Pandas, SciPy, scikit-learn, Matplotlib), SAS, SQL (SSMS), MATLAB, Power BI, R

**Quantitative Methods:** Stochastic Calculus · Monte Carlo Simulation · Finite Difference Methods · Dynamic Programming · Hamilton–Jacobi–Bellman Equations · Stochastic Optimal Control · Time-Series Analysis · Bayesian Methods · Regression Modeling · PCA / Factor Analysis

**Finance & Risk:** Derivatives Pricing (Black-Scholes, Binomial Trees, Monte Carlo) · Options Greeks · CVA · PD Modeling · Model Risk Management · Backtesting · Sharpe Ratio · Drawdown Analysis · Factor Models

## WORK EXPERIENCE

**FSRM Consultant — Model Risk** | **EY Canada** Montreal, QC | Jun 2024 – Present

- Administered a model risk management platform governing 800+ models across credit, market, and operational risk; engineered KPI dashboards in Power BI and optimized SQL data pipelines (SSMS) to automate inherent/residual risk scoring, reducing manual reporting time and enhancing regulatory oversight.
- Led wholesale credit risk PD model development for large corporate portfolios using a shadow rating methodology: performed feature engineering, statistical model validation, industry benchmarking, and regulatory alignment analysis in SAS and SQL, contributing to IFRS 9 / Basel-compliant risk assessments.
- Collaborated cross-functionally with risk managers, model owners, and senior stakeholders to design and implement model risk scoring methodology, ensuring consistency with SR 11-7 / E-23 model governance standards.

**Research Scholar — Stochastic Control** | **UQAM** Montreal, QC | Jan 2024 – Feb 2025

- Developed a dividend optimization framework for an insurance company under a mean-reverting Ornstein–Uhlenbeck surplus model with ruin-time penalization; formulated the problem as a stochastic optimal control problem and derived closed-form and numerical solutions via Hamilton–Jacobi–Bellman (HJB) equations — producing a Master's thesis.
- Implemented finite-difference numerical solvers and Monte Carlo simulations in Python (NumPy, SciPy) to validate theoretical results and perform sensitivity analysis across model parameters.

**Teaching Assistant — Actuarial Finance** | **UQAM** Montreal, QC | Jan 2024 – Apr 2024

- Delivered tutorials and graded assignments for Mathematics of Actuarial Finance II (ACT 5310), covering interest rate theory, life contingencies, and stochastic models in insurance and finance.

**Research Intern — Operations Research & Optimal Control** | **GERAD Lab** Montreal, QC | Feb 2022 – Jun 2022

- Modeled the optimal timing strategy for a retailer's transition to e-commerce as a stochastic impulse control problem; applied dynamic programming and optimal stopping theory to characterize switching thresholds under uncertainty.
- Implemented Monte Carlo simulations in Python to numerically analyze the effect of market popularity dynamics and data exploitation maturity on optimal transition timing.

**Trading Intern — Futures Strategies** | **Vifco** Tunisia | Jun 2021 – Aug 2021

- Researched and back-tested systematic trading strategies on futures markets using technical indicators (momentum, trend-following); iteratively optimized strategy parameters to improve Sharpe Ratio and Profit Factor, and produced performance attribution reports.

## QUANTITATIVE PROJECTS

**Option Pricing Library** | **Python, MATLAB**

- Implemented and benchmarked multiple derivatives pricing methods — Black-Scholes PDE (analytical & finite differences), binomial/trinomial trees, and Monte Carlo simulation — for European and American options; priced American options using Howard's policy-iteration algorithm, comparing convergence speed and numerical accuracy across methods.

**Equity Return Prediction** | **Python**

- Performed end-to-end quantitative research on Tunisian Stock Market data: applied Exploratory Data Analysis (EDA), PCA, Factor Analysis, and compared OLS, polynomial, and regularized regression models to forecast stock returns, evaluating out-of-sample  $R^2$ , RMSE, and residual diagnostics.